SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

Duration : 45 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I Sanjana.G(192511020) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Sanjana.G

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1) An intelligent agent is a system that observes its environment through sensors and performs actions using actuators to achieve a particular objective.

In a Smart home, Intelligent agents control lighting, temperature, and security based on user behaviours and environmental conditions.

(i) Simple Reflex Agent

A simple reflex agent takes action based only on the current input using condition-action rules.

Ex:

* If motion is detected, the light turns on.
* If the temperature is high, the fan turns on.
* If unauthorized entry is detected, the alarm activates.

It is fast and simple but cannot remember past information or learn user preferences.

(ii) Model-Based Agent

A model-based agent maintains an internal state of the environment using current and previous information.

Ex: It remembers room occupancy and prevents lights from turning off when a person is temporarily outside the sensor's range.

It handles changing environment effectively but requires additional memory.

(iii)

Goal - Based Agent

A goal - based agent selects actions to achieve a specific objective.

Ex: It adjusts the air conditioner to maintain the desired temperature and ensures that doors are locked when nobody is home.

It supports planning but may require more computation

(iv)

Utility - Based Agent

A utility - Based agent evaluate different outcomes and selects the most beneficial action based on factors such as comfort, energy consumption and security.

Ex: It adjust the air conditioner to maintain comfort while reducing electricity consumption.

It balances multiple objectives but requires a suitable utility function.

Most Effective Approach:

The model- based agent remembers environmental conditions, the goal - based agent works towards comfort and security, the utility - based agent balance energy efficiency and comfort and the simple reflex agent responds quickly to emergencies.

Conclusion:

A hybrid intelligent agent provides comfort, energy efficiency and security while adapting to user preferences and environmental changes.

2) Uninformed search strategies explore possible paths without prior knowledge of the maze's layout. Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help a robot find the exit.

(i) Uniform Cost Search (UCS)

UCS expands the node with the lowest total path cost using a priority queue.

Working:

1) Start from the Initial Position.
2) Explore available paths and Calculate their Costs
3) Select the node with the lowest Path Cost.
4) Repeat until the exit is reached.

Advantages:

- Finds the least-cost path when Step Costs are non-negative and suitable conditions hold.
- Handles different path costs effectively.

Disadvantages:

- Requires more memory
- May explore many unnecessary paths.
- Can be slow for large mazes.

(ii) Iterative Deepening Search

IDS repeatedly performs depth-limited searchs, increasing the depth limit each time until the Exit is found.

Working:

(i) Start with depth limit of 0.
(ii) Increase the limit to 1, 2, 3 and so on.
(iii) Repeat the search at each depth.
(iv) Stop when the exit is found.

Advantages:

- Uses less memory.
- Works well when the Solution depth is unknown.

Disadvantages:

- Repeats searches at shallow depths.
- May waste time exploring the same nodes.

4) Relationship with Intelligent Agents:

* Simple reflex agent
* Model-based agent
* Goal-based agent
* Utility-based agent

5) Most effective Combination

* UCS is preferred when path have different costs and the least-cost route is required.
* IDS is preferred when all movements have equal costs and memory is limited.

Conclusion: UCS is suitable for finding the least-cost path, while IDS is useful when memory is limited and the solution depth is unknown. Combining an appropriate search strategy with a model-based, goal-based agent improves decision-making and efficiency.